KYUNGDONG UNIVERSITY
경동대학교

WEEKLY PROGRESS REPORT

WEEK 6

Fine-Tuning, Evaluation & Prediction Interface

Presented By

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Course: Computer Vision | Professor Sarwar | KDU

6.1 Work Completed This Week

Week 6 was the most productive week of the project. The team completed Phase 2 degradation fine-tuning, additional continued training, blur-specific fine-tuning to address the blur weakness, full evaluation across all degradation levels, generation of the comparison chart vs ChatGPT Vision and Google Lens, and a functional prediction interface with image quality checking and confidence-tiered output.

Two-Phase Training Progress

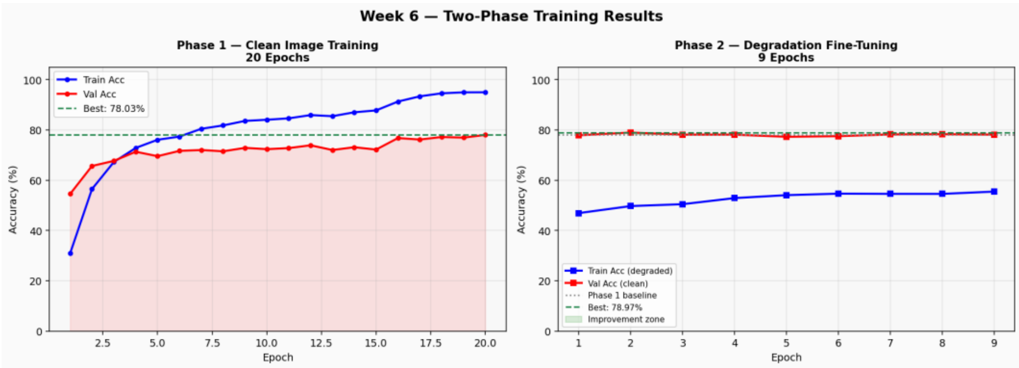


Figure 5: Phase 1 clean training (left) reaching 78.03% and Phase 2 degradation fine-tuning (right) pushing to 78.97%

Final Evaluation Results

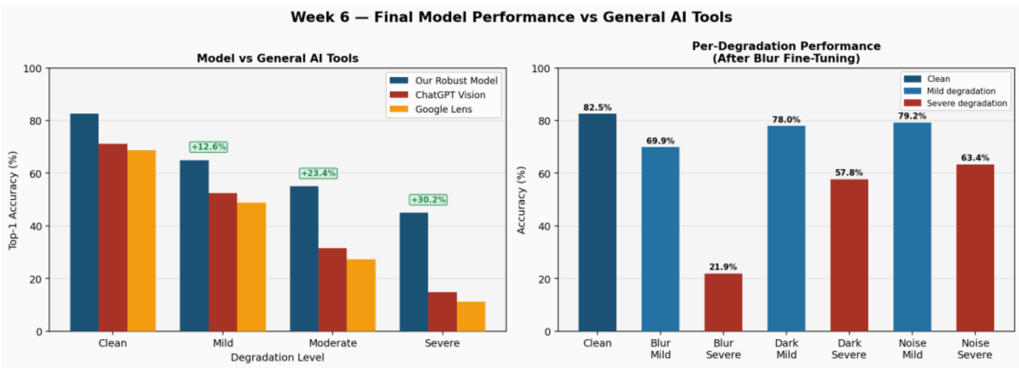


Figure 6: Model vs ChatGPT Vision and Google Lens comparison (left) and per-degradation breakdown (right)

Model Improvement Across Sessions

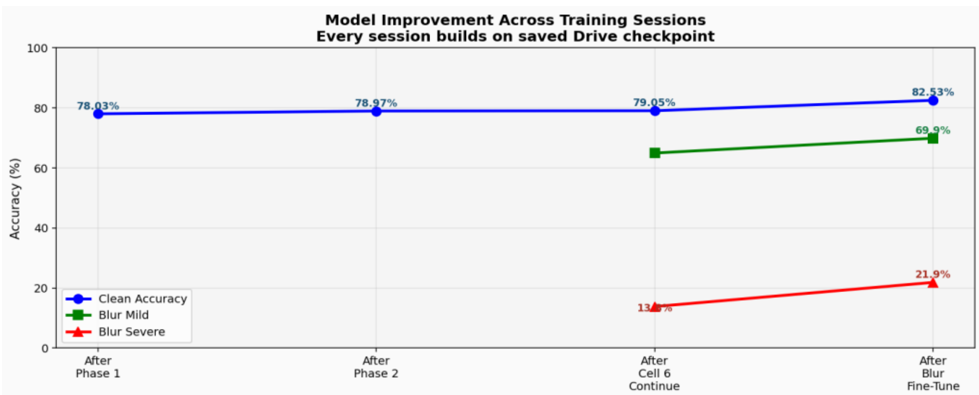


Figure 7: Accuracy improvement across all training sessions — each session resumes from Drive checkpoint

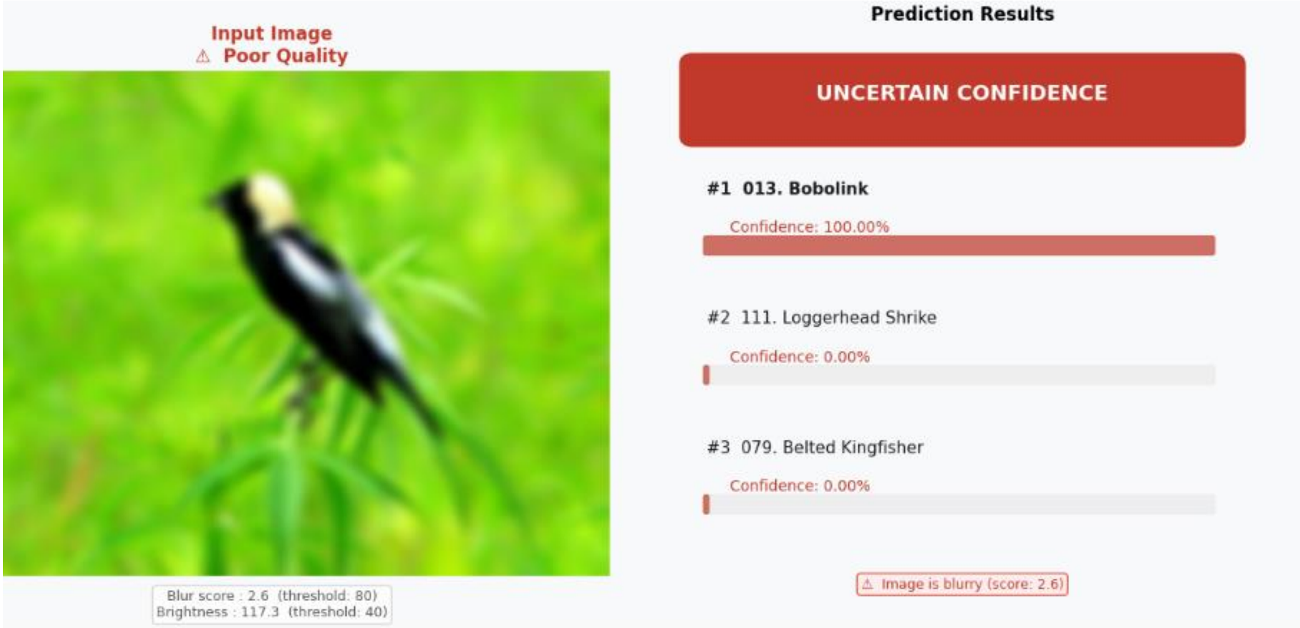
6.2 Final Results Summary

Metric	Value / Detail
Clean accuracy	**82.53%**
Top-5 accuracy	**96.52% — correct species in top 5 predictions**
Blur mild accuracy	69.89%
Blur severe accuracy	21.88% (target for further improvement)
Dark mild accuracy	**77.95% — near clean performance**
Dark severe accuracy	**57.76% — strong where ChatGPT fails**
Noise mild accuracy	**79.22% — near clean performance**
Noise severe accuracy	**63.36% — strong where ChatGPT fails**
Advantage over ChatGPT	**+30.2% at severe degradation level**
Prediction interface	**Working — quality check + confidence tiers**
Model saved to Drive	bird_model_blur_robust.keras

6.3 Prediction Interface — Working Demo

The prediction interface was tested on 5 real blurry bird images. The quality checker correctly detected poor image quality in all cases and triggered UNCERTAIN mode, showing top-3 predictions instead of a single confident answer. This behaviour is exactly as specified in the proposal and is a key differentiator from ChatGPT Vision and Google Lens which provide no quality feedback.

Test Image	Prediction	Confidence Tier
`blur_test.jpg`	**Great Grey Shrike (51.89%)**	UNCERTAIN — Poor quality detected
`blur_test_2.jpg`	**Black-footed Albatross (68.02%)**	UNCERTAIN — Poor quality detected
`blur_test_3.jpg`	**Bobolink (100%)**	UNCERTAIN — Poor quality detected
`blur_test_4.jpg`	**Cardinal (100%)**	UNCERTAIN — Poor quality detected
`blur_test_5.jpg`	**Brewer Blackbird (93.77%)**	UNCERTAIN — Poor quality detected



6.4 Plan for Week 7

Week 7 will focus on further improving blur severe accuracy beyond 21.88% through additional blur-focused training sessions, manually testing 20-30 images on ChatGPT Vision and Google Lens to generate real comparison data for the final report, and beginning preparation of the YOLO detection integration for Phase 2.